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Towards Interest Drift-driven User Representation Learning in Sequential Recommendation

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Abstract

Sequential recommendation (SR) aims to infer users' future interests and suggest the next items for them. Most SR methods learn one single vector to represent a user's recent interests, *i.e.*, a user representation. Despite their great success, most of them do not explicitly consider the phenomenon of users' *interest drift* when learning user representations. Moreover, interest drift presents two critical challenges for these SR methods: (1) how to explore the potential distributions of the users' varying interest drift levels; and (2) how to capture the interest drift-aware collaborative knowledge among the users. In this paper, we delve into the issue of interest drift in SR and propose a novel and generic framework, *i.e.*, **I**nterest **D**rift-driven **U**ser **R**epresentation **L**earning (IDURL), to enhance SR methods to tackle the above two challenges. Specifically, our IDURL contains an interest drift quantization (IDQ) module to enable a quantitative measurement of the interest drift. Moreover, a drift representation generation module models the users' latent varying levels of interest drift, and an interest drift-guided representation disentanglement module optimizes the distributions of the interest drift levels under the guidance of IDQ. Furthermore, an interest drift-aware representation alignment module helps to capture the interest drift-aware collaborative knowledge among users. Finally, the users' overall interest representations are obtained to calculate the preference scores on the candidate items. Extensive experiments on four public datasets show the effectiveness of our IDURL. The source code of our IDURL is available at: https://github.com/xiaolLIN/IDURL.

CCS Concepts

• **Information systems** → **Recommender systems**.

Keywords

Interest Drift, User Representation Learning, Sequential Recommendation

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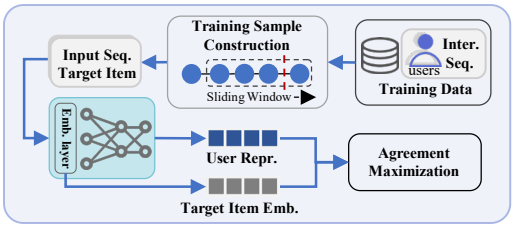


Figure 1: The traditional user representation learning paradigm adopted by most SR methods. This paradigm implicitly assumes that users' interests evolve stably.

1 Introduction

Sequential recommendation (SR) leverages the historical interactions to predict the next item for each user [12]. In SR, a vector is usually learned to represent each user's interests, *i.e.*, a user representation, which is then used to calculate the preference scores. To represent users' interests, deep learning-based techniques are prevalently applied in SR methods [16, 19, 47, 53, 65]. Based on the architecture of Transformer [50], some methods [39, 41, 55] design contrastive learning tasks to obtain better user representations. Recently, many research endeavors [28, 29, 51, 57] have integrated some side information to enhance user representation learning.

As illustrated in Figure 1, a user's interaction sequence in the training data can often be constructed into the input sequences and the corresponding target (next) items using a sliding window [63]. Many SR methods [16, 19, 65] based on item ID information and some side information-enhanced SR methods, such as NOVA [29], adopt a common paradigm of user representation learning (URL). They obtain a user representation by aggregating the embedding vectors transformed from an input sequence. Maximizing the agreement between the representation and the target item embedding during training, they optimize the user representation to approach the interest reflected by the target item. This facilitates predicting the expected items if the users' input sequences contain indicative items during the prediction stage.

Under the above paradigm, most SR methods implicitly assume that a user's interests evolve in a stable and deterministic manner [10, 11], and thus there is no significant discrepancy between

the interests reflected by the constructed input sequences and the corresponding target items. However, this assumption might not be true in all cases. In real scenarios, user interests may fluctuate substantially over time, which is known as ***interest drift*** [2, 59]. For example, as illustrated in Figure 2, user u_2 switches from sporting goods to electronics, and then to business style. Some SR methods [3, 23, 48] model the above multi-aspect interests in users' sequences, yet they still overlook the underlying reason for these phenomena, *i.e.*, interest drift. Although some recent methods [10, 11] have employed Gaussian distributions to model the instability of user interests, they only focus on the interest variations at the level of the dispersive item interactions, overlooking those between the target items and the entire input sequences. Moreover, we investigate the issue of interest drift and find that it presents the following two challenges for the SR methods under the above URL paradigm.

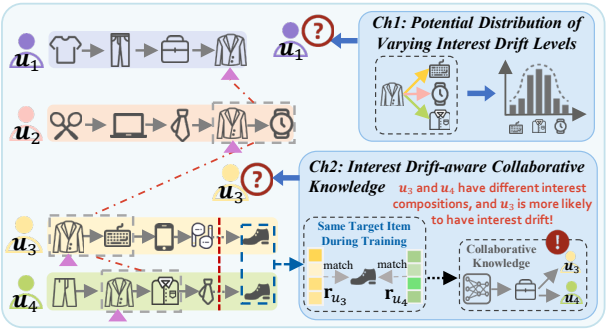


Figure 2: The SR methods adopting the traditional URL paradigm encounter difficulties in predicting the next items for user u_1 and user u_3, due to the corresponding challenges arising from the interest drift issue.

Ch1: Potential distribution of varying interest drift levels. In real scenarios, the subsequent items of a certain item might reflect the different levels of interest drift. As illustrated in Figure 2, a suit might be followed by a watch, a keyboard, or a shirt. Among these items, the keyboard reflects the most significant interest drift relative to the suit, followed by the watch, and the least is the shirt. The occurrence possibilities of varying levels of interest drift constitute the potential distribution of interest drift in a particular context, such as when the most recent item of user u_1 is a suit. Under the traditional URL paradigm, SR methods can force a user's representation to capture a specific level of interest drift reflected by the target item. However, they cannot simultaneously model the possibilities of the user's varying levels of interest drift and explore their potential distribution. Consequently, they might fail to infer the possible levels of interest drift for user u_1. Thus, exploring the potential distribution of the users' varying interest drift levels remains challenging.

Ch2: Interest drift-aware collaborative knowledge. During training, different users may be linked to a same target item, such as user u_3 and u_4 in Figure 2. Under the traditional paradigm, their respective representations are optimized to match the same target item embedding. In this way, SR methods capture some collaborative knowledge among these two users, that is, they might derive similar user representations and recommend similar items to users u_3 and u_4, such as a briefcase, during inference. However, u_3 and u_4 have different interest compositions, and they might also exhibit varying levels of interest drift at the next time step. The behavior history of u_3 indicates a high probability of interest drift, while u_4 is more likely to focus on business attire consistently. Thus, recommending a briefcase to u_3 may not be appropriate. And the oversight of these details will result in a suboptimal recommendation for u_3. Therefore, capturing the interest drift-aware collaborative knowledge among users remains a challenge.

To address the above challenges arising from the interest drift issue, we propose a generic framework, interest drift-driven user representation learning (IDURL), which can enhance the SR methods adopting the traditional URL paradigm. Our IDURL incorporates several well-designed key modules with distinct purposes. **To quantitatively measure the interest drift**, we propose an interest drift quantization (IDQ) module. In Figure 2, we qualitatively deem that the interest drift reflected by a watch is more significant than a shirt, which is based on external knowledge and may not be universally accepted. Therefore, we leverage multi-tag category information to quantize the interest drift between a user's input sequence and the target item. And the quantized results work well for the subsequent modules. **To address the first challenge**, we propose a drift representation generation (DRG) module and an interest drift-guided representation disentanglement (IDRD) module. In DRG, we generate various drift representations to model a user's potential varying levels of interest drift, based on the user's original representation derived from a backbone SR model. Moreover, in IDRD, we propose an additional learning objective to disentangle the specific drift representation from others under the guidance of IDQ results. This representation disentanglement module will help optimize the distribution of a user's potential interest drift levels during training. **To address the second challenge**, we propose an interest drift-aware representation alignment (IDRA) module. Based on the IDQ results, we establish a stricter criterion for identifying similar users to guarantee the same target items and the same interest drift degree. Then we align the sum pooling results of drift representations from the similar user pairs, as determined by the above criterion. In this way, IDRA explicitly draws the representations of similar users closer and captures the interest drift-aware collaborative knowledge. **For item prediction**, we take a candidate item embedding and a user's drift representations to concretize the interest drift distribution, by which we obtain the user's overall interest representation and the preference score on the item.

We summarize our main contributions as follows:

- We propose a novel and generic framework, *i.e.*, IDURL, which can enhance most SR methods to deal with the two challenges arising from the interest drift issue.
- We first propose a practical approach to quantitatively measure the interest drift between the input sequence and the target item. Then, we design two modules (*i.e.*, DRG and IDRD) to explore the potential distributions of users' varying interest drift levels in SR. Moreover, we design an IDRA module to capture the interest drift-aware collaborative knowledge among the users.
- We conduct extensive experiments on four public datasets, and verify the superiority of our IDURL as well as its strong compatibility across different backbone models.

2 Related Work

SR Methods Without the Consideration of the Interest Drift Issue. Early SR methods [14, 43] use Markov chains and matrix factorization to capture the sequential patterns. With the development of deep learning, some techniques are adopted in SR methods to represent the users' interests, such as RNNs [15, 16, 42], CNNs [47, 61], self-attention mechanism [19, 24, 26], MLPs [25, 65], and GNNs [4, 33]. As diffusion models (DMs) demonstrate their superior generative capabilities in computer vision [17, 44] and multi-modal generation [9, 20], some methods [34, 54, 58] have utilized DMs in SR and achieved competitive performance. Some later methods employ contrastive learning to obtain better representations, where they construct self-supervision signals with data-level augmentation [5, 31, 55] and model-level augmentation [6, 39, 41], or their combination [39, 40, 66]. Some recent methods capture user interests via a multi-interest learning strategy [3, 23, 48, 56], as well as disentangled learning techniques [8, 62, 64]. Another line of research endeavors [28–30, 51, 57] considers additional side information to enhance user representation learning. However, these methods do not well account for the issue of interest drift, leaving room for more accurate predictions of users' future interests.

Interest Drift-aware Recommender Systems. Early works on recommender systems study the issue of interest drift based on some heuristic approaches, such as item similarity calculation [2] and clustering algorithms [27]. Some works explore this issue in some specific task scenarios, such as news recommendation [37] and point-of-interest recommendation [7, 32, 59]. In SR, PPD [45] and its end-to-end version [35] split a user's sequence into several subsequences by detecting the interest drift points, and then only consider the subsequences containing similar interest patterns to the last one. PERIS [18] proposes a task to predict whether users' interests will be sustained beyond the training time. Some methods [10, 11] model the uncertainty of SR with Gaussian distributions. DT4SR [10] represents each item with a Gaussian distribution and adopts two separate Transformers to model the mean and covariance embeddings, respectively. Then, it uses Wasserstein distance to measure the distances between the inferred distribution and those of the candidate items. Based on DT4SR, STOSA [11] further devises a Wasserstein attention mechanism to capture the collaborative transitivity between items. However, these methods have not yet addressed the challenges posed by the interest drift issue.

3 Problem Formulation

Notations. We have a set of users $\mathcal{U}=\{u_1, u_2 \ldots, u_{|\mathcal{U}|}\}$, and a set of items $\mathcal{V}=\{v_1, v_2 \ldots, v_{|\mathcal{V}|}\}$. Since we aim to leverage multi-tag category information to measure users' interest drift, we consider a set of item categories $C = \{c_1, c_2 \ldots, c_{|C|}\}$. Each item $v_i \in \mathcal{V}$ contains multi-tag category information $C_{v_i} \subset C$. For a user $u \in \mathcal{U}$, his or her input sequence is defined as $\mathcal{S}_u = \left\{s_1^u, \ldots, s_t^u, \ldots, s_n^u\right\}$, where n is the sequence length. Each interaction $s_t^u = (v_t^u, C_{v_t^u})$ denotes that user u interacts with item v_t^u at time step t and item v_t^u is associated with a set of categories $C_{v_t^u}$.

Task. Given the input sequence $\mathcal{S}_u$, the task is to predict the next item v_{n+1}^u with the highest probability for user u:

$$v_{n+1}^u = \arg\max_{v_i \in \mathcal{V}} P\left(v_i \mid \mathcal{S}_u\right). \tag{1}$$

4 Methodology

We propose a novel and generic framework, IDURL, to address the aforementioned challenges arising from the interest drift issue. Our IDURL can be combined with most SR models and boost their performance. The pipeline of our IDURL that takes SASRec [19] as an example backbone model is shown in Figure 3. We will elaborate on the details of our IDURL according to the pipeline.

4.1 Interest Drift Quantization (IDQ)

To effectively investigate the interest drift, we need a metric for the quantitative measurement of such an abstract concept. The category information of items can effectively reflect the users' fine-grained interests [1], so we propose to leverage this information to provide a concrete quantization of interest drift. Specifically, we first formally define interest drift magnitude (IDM), and design a bucketing function to map IDM to a discrete drift degree.

Interest Drift Magnitude. During training, a user's interaction sequence can be constructed into several pairs of the input sequence and the corresponding target items using a sliding window [63]. We quantify the interest dependency strength between an input sequence $\mathcal{S}_u$ and its target item v_{n+1}^u through the overlap ratio of the categories. Then we transform the result to the interest drift magnitude (IDM). Formally, IDM is defined as follows:

$$\mathrm{IDM}\left(\mathcal{S}_u, v_{n+1}^u\right) = 1 - \frac{\left|C_{v_{n+1}^u} \cap \bigcup_{t=1}^{n} C_{v_t^u}\right|}{\left|C_{v_u^{n+1}}\right|}, \tag{2}$$

where $C_{v_{n+1}^u}$ denotes the category set associated with v_{n+1}^u, and $\bigcup_{t=1}^{n} C_{v_t^u}$ is the union of the category sets of the input sequence. We can see that the second term of IDM is the overlap ratio between the categories of the target item and those within the input sequence. Hence, a lower overlap ratio implies a higher IDM. Intuitively, a user's interest has been entirely changed if the categories of the target item are not present in the input sequence at all. Note that IDM ranges from zero to one.

Interest Drift Discretization. We aim to explicitly model users' potential various levels of interest drift. Thus, we design a bucketing function to map IDMs to K distinct drift degrees, *i.e.*, $\{1, 2, \ldots, K\}$. For an input IDM value x, the function is formulated as follows:

$$q = \mathrm{bucket}(x) = \begin{cases} 1, & \text{if } x = 0 \\ \lceil (K-2) \cdot x \rceil + 1, & \text{if } 0 < x < 1 \\ K, & \text{if } x = 1 \end{cases}. \tag{3}$$

The cases where IDMs are zero and one indicate two scenarios, *i.e.*, no interest drift and complete interest drift, which are mapped to 1 and K, respectively. For the remaining range between zero and one, where IDMs signify the partial drift of users' interests, we evenly divide the range into $K-2$ segments and map them to the drift degrees $\{2, \ldots, K-1\}$. For brevity, we formulate the overall interest drift quantization process as $q = \mathrm{IDQ}(\mathcal{S}_u, v_{n+1}^u)$, which is illustrated in Figure 3(b).

4.2 User Representation Encoding

In our IDURL, we employ the backbone model of an existing SR method to obtain the user representation based on the input sequence. For a typical SR method based on the ID information such

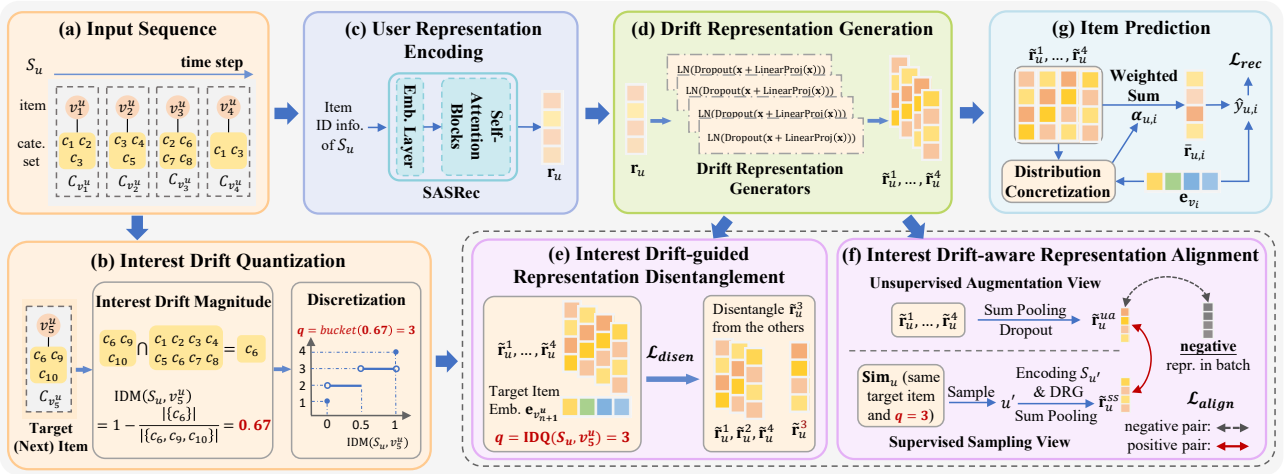


Figure 3: The pipeline of our IDURL with $K = 4$ and SASRec as an example backbone model: (a~b) Our IDURL leverages the multi-tag category information to quantize the interest drift between the input sequence and the target item during training. (c) The item ID information of the input sequence is encoded as the user representation by the backbone model. (d) The drift representations are generated to model the user's potential varying levels of interest drift. (e~f) During training, a drift representation is disentangled from the others, under the guidance of quantized interest drift degree, *i.e.*, $q = 3$. And the sum-pooling result of these representations is aligned with that of an interest drift-aware similar user. (g) Finally, a candidate item embedding is used to obtain the user's overall interest representation and calculate the preference score for the item.

as SASRec [19], we only leverage the item IDs within the input sequence, which is illustrated in Figure 3(c). Note that our IDURL is highly compatible with most SR methods that adopt the paradigm in Figure 1. Therefore, by adding the category information and replacing the backbone model, a side information-enhanced SR method, e.g., NOVA [29], can also be combined with our IDURL.

An input sequence is embedded as low-dimensional dense vectors, which are then inputted into a sequential encoder to obtain the user representation. We formulate the overall process as follows:

$$\mathbf{r}_u = \mathbf{URE}\left(\mathcal{S}_u; \mathcal{M}\right), \tag{4}$$

where $\mathbf{URE}(\cdot)$ denotes the user representation encoding process using an SR backbone model $\mathcal{M}$, and $\mathbf{r}_u \in \mathbb{R}^d$ is the representation of user u. Note that d is the latent dimensionality.

4.3 Drift Representation Generation (DRG)

Most SR methods do not explicitly consider the interest drift issue and represent a user based on an interest-stable assumption [11]. However, the dynamic nature of user interests often leads to the phenomenon of interest drift. Although the SR methods with the traditional URL paradigm can optimize the user representation to approximate the interest w.r.t. the target item, they fail to model the user's varying levels of interest drift synchronously.

To model the user's drifted interest, we design a drift representation generator (DRG), which takes the original user representation $\mathbf{r}_u$ to generate the drift representation (DR):

$$\tilde{\mathbf{r}}_u = \mathrm{DRG}\left(\mathbf{r}_u\right), \tag{5}$$

$$\mathrm{DRG}\left(\mathbf{x}\right) = \mathrm{LN}\left(\mathrm{Dropout}\left(\mathbf{x} + \mathrm{LinearProj}\left(\mathbf{x}\right)\right)\right), \tag{6}$$

$$\mathrm{LinearProj}\left(\mathbf{x}\right) = \mathbf{W}\mathbf{x} + \mathbf{b}, \tag{7}$$

where $\mathrm{LinearProj}(\cdot)$ is a linear projection used to introduce the semantic information that is different from $\mathbf{r}_u$ [60], and the residual connection [13], *i.e.*, $\mathbf{x} + \mathrm{LinearProj}\left(\mathbf{x}\right)$, allows the new representation to preserve the essential features of the original one when incorporating the projection result. $\mathrm{LN}(\cdot)$ (layer normalization) and $\mathrm{Dropout}(\cdot)$ are used to improve training stability and avoid overfitting, respectively.

To model the user's K potential levels of interest drift, we employ K different DRGs to obtain distinct DRs, *i.e.*, $\{\tilde{\mathbf{r}}_u^1, \ldots, \tilde{\mathbf{r}}_u^k, \ldots, \tilde{\mathbf{r}}_u^K\}$ for a same representation $\mathbf{r}_u$, where $\tilde{\mathbf{r}}_u^k = \mathrm{DRG}_k\left(\mathbf{r}_u\right) \in \mathbb{R}^d$. These new representations will contain diverse semantic information [60]. Under the effect of the subsequent disentanglement module, each of these DRs will capture a specific level of interest drift.

4.4 Interest Drift-guided Representation Disentanglement (IDRD)

During training, our IDRD aims to explore users' potential distributions of varying levels of interest drift. For user u, we employ the normalized attention weights between the target item embedding and the generated DRs to concretize his or her interest drift distribution: $\mathrm{softmax}([\tilde{\mathbf{r}}_u^1 \cdot \mathbf{e}_{v_{n+1}^u}, \ \ldots, \ \tilde{\mathbf{r}}_u^K \cdot \mathbf{e}_{v_{n+1}^u}])$.

However, because the generated DRs of $\mathbf{r}_u$ lack the guidance for disentangled representation learning, they might easily collapse into similar ones [8], leading to distinguishable distribution weights. Thus, we propose to strengthen the representation disentanglement under the guidance of the supervised signal from the IDQ module. Specifically, we obtain a quantized interest drift degree q, *i.e.*, $q = \mathrm{IDQ}(\mathcal{S}_u, v_{n+1}^u)$. Then, we take the target item embedding as an anchor to pull the DR specified by value q closer and push other DRs away via a disentanglement objective:

$$\mathcal{L}_{disen} = -\sum_{(\mathcal{S}_u, v_{n+1}^u) \in \mathcal{D}} \log \frac{\exp\left(\tilde{\mathbf{r}}_u^q \cdot \mathbf{e}_{v_{n+1}^u}\right)}{\sum_{j=1}^{K} \exp\left(\tilde{\mathbf{r}}_u^j \cdot \mathbf{e}_{v_{n+1}^u}\right)}, \tag{8}$$

where $\mathcal{D}$ denotes the training data. Leveraging the supervision of the quantized interest drift degree q and the semantic information of the target item embedding $\mathbf{e}_{v^u_{n+1}} \in \mathbb{R}^d$, we disentangle $\tilde{\mathbf{r}}^q_u$ from other DRs and enable it to capture the expected interest drift level reflected by the target item. By minimizing Eq.(8), the DRG module will be gradually trained to generate the DRs that capture the distinct levels of interest drift. Moreover, the distributions about the users' potential varying levels of interest drift are also optimized, within the context specified by the input sequence and the corresponding target item. In the end, the concretized interest drift distributions are conducive to representing the users' interest strengths with high confidence in the prediction stage.

4.5 Interest Drift-aware Representation Alignment (IDRA)

A same target item might reflect vastly different levels of interest drift relative to distinct users' input sequences, and these users may not necessarily be similar in terms of their interest compositions either. However, most SR methods with the traditional URL paradigm overlook these details. This results in these SR methods' inability to capture the interest drift-aware collaborative knowledge and tailor the optimal recommendation results. To alleviate this problem, we propose an interest drift-aware representation alignment module.

Firstly, we establish a criterion to identify interest drift-aware similar users, *i.e.*, the users who not only share the same target item but also have the same interest drift degree between the target item and their input sequences. The same drift degree means the similar IDM values and category overlap ratios, which helps to bring similar interest compositions within the users' input sequences. For user u with drift degree $q = \mathrm{IDQ}(\mathcal{S}_u, v^u_{n+1})$, his or her similar user set is defined as follows:

$$\mathbf{Sim}_u = \left\{ u' \,\middle|\, u' \in \mathcal{U}, v^{u'}_{n+1} = v^u_{n+1}, \mathrm{IDQ}(\mathcal{S}_{u'}, v^{u'}_{n+1}) = q \right\}. \tag{9}$$

Then we sample a similar user u' from $\mathbf{Sim}_u$. We utilize the shared backbone model and DRG module to obtain the drift representations of user u', *i.e.*, $\{\tilde{\mathbf{r}}^1_{u'}, \ldots, \tilde{\mathbf{r}}^K_{u'}\}$. Because we aim to realize the alignment in terms of the user's overall interest composition, we take the sum pooling of these representations as the basic views (bv) of user u and user u':

$$\tilde{\mathbf{r}}^{bv}_u = \sum\nolimits_{k=1}^{K} \tilde{\mathbf{r}}^k_u, \ \tilde{\mathbf{r}}^{bv}_{u'} = \sum\nolimits_{k=1}^{K} \tilde{\mathbf{r}}^k_{u'}. \tag{10}$$

Following [41], we construct two contrastive views in a contrastive learning task, including the unsupervised augmentation (ua) view and the supervised sampling (ss) view. Specifically, we save the computation cost and directly conduct a simple model-level augmentation, *i.e.*, Dropout [46], on the basic view of user u to acquire the view ua. To align user u with an interest drift-aware similar user, we take the basic view of the sampled similar user u' as the view ss. Two contrastive views can be respectively formulated as: $\tilde{\mathbf{r}}^{ua}_u = \mathrm{Dropout}(\tilde{\mathbf{r}}^{bv}_u)$, $\tilde{\mathbf{r}}^{ss}_u = \tilde{\mathbf{r}}^{bv}_{u'}$.

Subsequently, we treat the contrastive views of a same user as a positive pair and those from two different users as negative pairs, and adopt InfoNCE [49] as the contrastive loss function. Finally, the learning objective of representation alignment can be formulated as follows:

$$\mathcal{L}_{align} = \sum_{u \in \mathcal{U}} \mathcal{L}_{cl}(\tilde{\mathbf{r}}^{ua}_u, \tilde{\mathbf{r}}^{ss}_u) + \mathcal{L}_{cl}(\tilde{\mathbf{r}}^{ss}_u, \tilde{\mathbf{r}}^{ua}_u), \tag{11}$$

$$\mathcal{L}_{cl}(\mathbf{x}^1_u, \mathbf{x}^2_u) = -\log \frac{\exp\left(\mathrm{s}(\mathbf{x}^1_u, \mathbf{x}^2_u)\right)}{\exp\left(\mathrm{s}(\mathbf{x}^1_u, \mathbf{x}^2_u)\right) + \sum\limits_{u^- \in \mathcal{N}} \exp\left(\mathrm{s}(\mathbf{x}^1_u, \mathbf{x}^2_{u^-})\right)} \tag{12}$$

$$\mathrm{s}(\mathbf{x}^1_u, \mathbf{x}^2_u) = \mathrm{LN}(\mathbf{x}^1_u) \cdot \mathrm{LN}(\mathbf{x}^2_u), \tag{13}$$

where $(\mathbf{x}^1_u, \mathbf{x}^2_u)$ is an aligned positive pair for user u, $\mathcal{N}$ indicates the set of other negative users from the same training batch, and $\mathrm{s}(\cdot,\cdot)$ denotes the inner product of two normalized representations. By minimizing the alignment loss, *i.e.*, Eq.(11), we explicitly draw interest drift-aware similar users closer. In this way, the interest drift-aware collaborative knowledge among users is well captured when learning user representations.

4.6 Item Prediction and Training

Prediction via interest drift distribution concretization (IDDC). Based on the drift representations that capture a user's varying levels of interest drift, we take a candidate item embedding $\mathbf{e}_{v_i} \in \mathbb{R}^d$ to concretize the user's potential interest drift distribution, which can be formulated as follows:

$$\boldsymbol{\alpha}_{u,i} = \mathrm{softmax}\left(\left[\tilde{\mathbf{r}}^1_u \cdot \mathbf{e}_{v_i}, \ \ldots, \ \tilde{\mathbf{r}}^K_u \cdot \mathbf{e}_{v_i}\right]\right), \tag{14}$$

where $\boldsymbol{\alpha}_{u,i} = \left[\boldsymbol{\alpha}^1_{u,i}, \ldots, \boldsymbol{\alpha}^K_{u,i}\right] \in \mathbb{R}^K$. With such a distribution, we attain the overall interest representation of user u about item v_i, and then calculate the preference score on this item as follows:

$$\bar{\mathbf{r}}_{u,i} = \sum_{k=1}^{K} \boldsymbol{\alpha}^k_{u,i} \cdot \tilde{\mathbf{r}}^k_u, \tag{15}$$

$$\hat{y}_{u,i} = f(u, v_i) = \bar{\mathbf{r}}_{u,i} \cdot \mathbf{e}_{v_i}. \tag{16}$$

Taking into account the concretized distribution, the interest representation $\bar{\mathbf{r}}_{u,i}$ can represent the user's interest strength w.r.t. item v_i with high confidence. Note that our prediction approach (*i.e.*, Eq.(14-16)) is implemented in a way that parallelizes all the candidate items.

During the prediction stage, the pipeline of our IDURL is $(a) \rightarrow (c) \rightarrow (d) \rightarrow (g)$, *i.e.*, the first row of modules in Figure 3. Our IDURL uses the backbone model and DRG module to obtain the drift representations for user u, and then calculates the preference scores. By ranking the scores of all items, the one with the highest score is recommended as the next item for user u:

$$v^u_{n+1} = \arg\max_{v_i \in \mathcal{V}} f(u, v_i). \tag{17}$$

Multi-task training. For user u, we attain the probability distribution of all the candidate items: $\hat{\mathbf{y}}_u = \mathrm{softmax}(\hat{y}_{u,1}, \ldots, \hat{y}_{u,|\mathcal{V}|}) \in \mathbb{R}^{|\mathcal{V}|}$. Then we adopt the cross-entropy loss as the optimization objective for the recommendation task:

$$\mathcal{L}_{rec} = -\sum_{u \in \mathcal{U}} \sum_{i=1}^{|\mathcal{V}|} \mathbf{y}_u[i] \log(\hat{\mathbf{y}}_u[i]), \tag{18}$$

where $\mathbf{y}_u$ is the ground truth (*i.e.*, a one-hot vector) w.r.t. user u.

We adopt the multi-task learning strategy to train our IDURL. And the overall objective combines the item recommendation, disentanglement, and alignment losses:

$$\mathcal{L} = \mathcal{L}_{rec} + \lambda_1 \mathcal{L}_{disen} + \lambda_2 \mathcal{L}_{align}, \tag{19}$$

where λ_1, λ_2 are the weight parameters to balance the corresponding loss terms.

4.7 Time Complexity Analysis

During training, the extra time cost of our IDURL comes from our representation alignment module, *i.e.*, IDRA. However, the similar user set is predetermined before training and does not increase any complexity. Compared with DuoRec [41] and MCLRec [39] that need to encode the sequences for both contrastive views, our IDRA directly conducts dropout operation to get the view ua, which reduces the complexity by half.

During inference, IDRA is no longer needed. We take SASRec as the backbone model to analyze the time complexity of our IDURL. As stated in [19], SASRec takes $O(n^2d + nd^2)$ to obtain $\mathbf{r}_u$, and $O(|\mathcal{V}|d)$ to predict scores, *i.e.*, $\hat{\mathbf{y}}_u = softmax(\mathbf{r}_u\mathbf{E}^\top)$, where $\mathbf{E} \in \mathbb{R}^{|\mathcal{V}|\times d}$ is the item embedding table. So, the complexity of SASRec is $O(n^2d+nd^2+|\mathcal{V}|d)$. As for our IDURL, it takes $O(Kd^2)$ to obtain K drift representations (DRs), and $O((Kd+K)|\mathcal{V}|)$ to conduct item prediction with IDDC. Thus, the time complexity of our IDURL is $O(n^2d + nd^2 + Kd^2 + (Kd + K)|\mathcal{V}|)$. Since K is relatively small (e.g., 4) and the item prediction module is parallelly implemented for GPU acceleration, our IDURL costs a little more in inference than SASRec. In practice, we can also use the averaged results of DRs to predict the scores more efficiently, *i.e.*, $\hat{y}_{u,i} = \frac{1}{K}\sum_1^K \tilde{\mathbf{r}}_u^k \cdot \mathbf{e}_{v_i}$, although the performance may be slightly worse.

5 Experiments

In this section, we conduct extensive experiments to answer the following research questions:

- **RQ1**: Can our IDURL be equipped on different backbone models and effectively improve their performance?
- **RQ2**: How does our IDURL perform compared with the state-of-the-art SR methods?
- **RQ3**: What is the impact of distinct modules in our IDURL?
- **RQ4**: How do some key hyperparameters affect the performance of our IDURL?
- **RQ5**: Why can our IDURL achieve superior performance?
- **RQ6**: How is the efficiency of our IDURL in the inference stage compared with the baselines?

Table 1: Statistics of the four processed datasets. Note that "# Avg. Tags / Item" means the average number of category tags associated with each item.

Dataset	Beauty	Sports	Toys	Yelp
# Users	22,363	35,598	19,412	30,499
# Items	12,101	18,357	11,924	20,068
# Categories	237	1,470	466	1,001
# Avg. Tags / Item	3.13	3.95	2.30	4.84
# Avg. Interactions / User	8.9	8.3	8.6	10.4
# Avg. Interactions / Item	16.4	16.1	14.1	15.8
# Interactions	198,502	296,337	167,597	317,182
Sparsity	99.93%	99.95%	99.93%	99.95%

5.1 Experimental Setup

5.1.1 ***Datasets.*** We conduct comprehensive experiments on four commonly used datasets, *i.e.*, Beauty, Sports, Toys, and Yelp. Specifically, **Beauty**, **Sports** and **Toys** are three subcategories of the Amazon review datasets[1] [36]. **Yelp** is a widely used business recommendation dataset collected by the homonymic platform[2]. These four datasets all contain multi-tag item category information. For each raw dataset, we follow [19, 57, 65] and remove the users and items that occur less than five times. The statistics of the four processed datasets are summarized in Table 1.

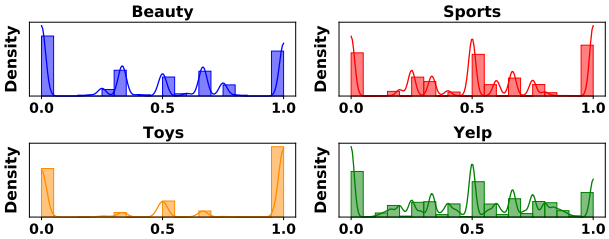


Figure 4: The IDM distributions of the training samples in the four processed datasets.

For dataset partition, we follow [19, 57, 65] and adopt the leave-one-out principle to split each user's interaction sequence into three parts, *i.e.*, the last interaction for test, the penultimate one for validation, and the remaining part for training. Then, for training data, we use the default setting of RecBole [63] and construct training samples (the pairs of the input sequence and target item) through a sliding window to fully leverage the temporal information in the sequences. Finally, we show the interest drift magnitude (*i.e.*, Eq.(2)) distributions of the training samples of the processed datasets in Figure 4, which confirms the existence of the phenomenon of varying levels of interest drift in the datasets.

5.1.2 ***Evaluation Metrics.*** We take Recall@k and NDCG@k as the evaluation metrics, where $k \in \{10, 20\}$. Following the suggestions in [22], we rank the preference scores on *all the items* rather than on only some sampled ones for a more reasonable comparison.

5.1.3 ***Baselines.*** Our IDURL is a generic framework that can be combined with most SR methods. To verify its compatibility, we combine our IDURL with different representative SR methods and investigate their performance compared with that of their original versions. Specifically, we choose **GRU4Rec** [16], **Caser** [47], **SASRec** [19] and **FMLP-Rec** [65] as the backbone models, which stand for four types of representative SR models based on RNN, CNN, Transformer and MLP, respectively.

To verify the superiority of our IDURL, four groups of baselines are compared in our experiments. The first group contains two interest drift-aware SR methods, including **DT4SR** [10] and **STOSA** [11]. The second group consists of two multi-interest SR methods, including **ComiRec** [3] and **ComiRec+** (a variant that is trained with our disentanglement loss, *i.e.*, Eq.(8), to bring more uniqueness for the generated multiple user representations). The third group comprises three contrastive learning-based SR methods, including **DuoRec** [41], **DuoRec+** and **MCLRec** [39]. Note that

[1] http://jmcauley.ucsd.edu/data/amazon/
[2] https://www.yelp.com/dataset

Table 2: The overall performance comparison (with Recall as R and NDCG as N) on four datasets, where the best and runner-up results of each column are marked in bold and underlined, respectively. * indicates a significance level of $p < 0.01$ based on a two-sampled t-test between the backbone models and their corresponding enhanced versions equipped with our IDURL.

Dataset	Beauty				Sports				Toys				Yelp			
Metrics	R@10	N@10	R@20	N@20	R@10	N@10	R@20	N@20	R@10	N@10	R@20	N@20	R@10	N@10	R@20	N@20
GRU4Rec	0.0598	0.0337	0.0875	0.0407	0.0319	0.0164	0.0483	0.0208	0.0499	0.0279	0.0732	0.0337	0.0399	0.0202	0.0667	0.0271
w/ IDURL	0.0768*	0.0446*	0.1011*	0.0516*	0.0394*	0.0208*	0.0619*	0.0267*	0.0601*	0.0367*	0.0951*	0.0441*	0.0517*	0.0241*	0.0838*	0.0342*
Improv.	28.43%	32.34%	15.54%	26.78%	23.51%	26.83%	28.16%	28.37%	20.44%	31.54%	29.92%	30.86%	29.57%	19.31%	25.64%	26.20%
Caser	0.0497	0.0261	0.0783	0.0333	0.0284	0.0143	0.0444	0.0183	0.0373	0.0197	0.0566	0.0245	0.0367	0.0191	0.0561	0.0242
w/ IDURL	0.0587*	0.0308*	0.0903*	0.0387*	0.0315*	0.0161*	0.0514*	0.0211*	0.0491*	0.0245*	0.0704*	0.0309*	0.0411*	0.0223*	0.0641*	0.0281*
Improv.	18.11%	18.01%	15.33%	16.22%	10.92%	12.59%	15.77%	15.30%	31.64%	24.37%	24.38%	26.12%	11.99%	16.75%	14.26%	16.12%
SASRec	0.0859	0.0423	0.1197	0.0508	0.0494	0.0228	0.0711	0.0283	0.0924	0.0448	0.1254	0.0531	0.0639	0.0397	0.0916	0.0466
w/ IDURL	**0.0935***	**0.0482***	0.1328*	**0.0581***	0.0542*	0.0266*	0.0796*	0.0333*	0.1012*	0.0503*	0.1402*	0.0599*	**0.0726***	0.0422*	**0.1072***	0.0509*
Improv.	8.85%	13.95%	10.94%	14.37%	9.72%	16.67%	11.95%	17.67%	9.52%	12.28%	11.80%	12.81%	13.62%	6.30%	17.03%	9.23%
FMLP-Rec	0.0873	0.0432	0.1233	0.0525	0.0504	0.0236	0.0733	0.0298	0.0939	0.0448	0.1287	0.0540	0.0645	0.0405	0.0932	0.0473
w/ IDURL	**0.0935***	0.0467*	**0.1348***	0.0573*	0.0552*	**0.0271***	0.0809*	**0.0345***	0.1027*	0.0508*	0.1404*	**0.0605***	0.0724*	**0.0427***	0.1014*	**0.0515***
Improv.	7.10%	8.10%	9.33%	9.14%	9.52%	14.83%	10.37%	15.77%	9.37%	13.39%	9.09%	12.04%	12.25%	5.43%	8.80%	8.88%
NOVA	0.0860	0.0436	0.1235	0.0531	0.0511	0.0239	0.0744	0.0297	0.0937	0.0494	0.1275	0.0579	0.0659	0.0406	0.0945	0.0478
w/ IDURL	0.0914*	0.0473*	0.1316*	0.0574*	**0.0565***	0.0270*	**0.0824***	0.0335*	**0.1034***	**0.0509***	**0.1410***	0.0604*	0.0701*	0.0420*	0.1034*	0.0504*
Improv.	6.28%	8.49%	6.56%	8.10%	10.57%	12.97%	10.75%	12.79%	10.35%	3.04%	10.59%	4.32%	6.37%	3.45%	9.42%	5.44%
DT4SR	0.0781	0.0377	0.1114	0.0461	0.0431	0.0198	0.0635	0.0248	0.0889	0.0414	0.1193	0.0496	0.0571	0.0371	0.0816	0.0432
STOSA	0.0834	0.0429	0.1166	0.0506	0.0459	0.0213	0.0662	0.0264	0.0939	0.0456	0.1258	0.0545	0.0598	0.0380	0.0858	0.0445
ComiRec	0.0765	0.0374	0.1061	0.0449	0.0443	0.0204	0.0641	0.0255	0.0819	0.0393	0.1091	0.0462	0.0567	0.0365	0.0801	0.0424
ComiRec+	0.0786	0.0392	0.1121	0.0468	0.0467	0.0216	0.0661	0.0265	0.0871	0.0418	0.1182	0.0497	0.0584	0.0371	0.0843	0.0435
DuoRec	0.0876	0.0448	0.1249	0.0545	0.0517	0.0241	0.0759	0.0304	0.0968	0.0465	0.1352	0.0542	0.0656	0.0397	0.0978	0.0477
DuoRec+	0.0883	0.0453	0.1254	0.0557	0.0525	0.0252	0.0767	0.0307	0.0978	0.0472	0.1367	0.0561	0.0685	0.0402	0.0976	0.0473
MCLRec	0.0874	0.0452	0.1256	0.0548	0.0495	0.0239	0.0712	0.0292	0.0934	0.0465	0.1321	0.0558	0.0684	0.0404	0.1012	0.0486
DIF-SR	0.0868	0.0447	0.1223	0.0537	0.0517	0.0239	0.0753	0.0298	0.0948	0.0492	0.1285	0.0574	0.0661	0.0405	0.0975	0.0484
MSSR	0.0889	0.0452	0.1262	0.0544	0.0510	0.0240	0.0751	0.0305	0.0942	0.0474	0.1281	0.0554	0.0693	0.0417	0.1006	0.0495
ASIF	0.0879	0.0455	0.1251	0.0541	0.0519	0.0242	0.0761	0.0301	0.0962	0.0463	0.1313	0.0552	0.0691	0.0415	0.1002	0.0493

DuoRec+ is an enhanced version of DuoRec, which further satisfies the requirement of the same interest drift degree in Eq.(9) during the supervised sampling process. Our IDURL leverages multi-tag category information to enhance user representation learning (URL). To highlight the effectiveness of our method in using category information, the last group of baselines is about side-information integrated SR methods, including **NOVA** [29], **DIF-SR** [57], **MSSR** [28] and **ASIF** [51]. As mentioned in Section 4.2, the URL paradigm of NOVA is the same as that of the above four representative SR methods, for which we also take it as a backbone model of our IDURL.

5.1.4 ***Implementation Details.*** We implement and evaluate all the baselines and our IDURL based on RecBole [63] for a fair performance comparison. For the last group of baselines, we follow the implementation of DIF-SR [57] to obtain the sum-pooling result of the multi-tag category embeddings associated with each item. This result is treated as a unified category embedding of the corresponding item, which is then input into the models to fuse the category information. For all the methods, we train them with the same optimizer (*i.e.*, Adam [21]), learning rate (*i.e.*, 1e-4), and batch size (*i.e.*, 256). Since the cross-entropy loss shows its superiority over its binary counterpart [38], we adopt it as the loss function of the main recommendation tasks for all methods. For the latent dimensionality d of all the methods, we choose its value from $\{64, 128\}$. As for the sequence length n, we follow the default setting of RecBole and set it as 50, which is also the sliding window size when constructing the training samples. We tune the other hyperparameters of each baseline according to the suggestions in the original papers. For the configurations of our IDURL, we choose λ_1 from $\{0.1, 0.3, 0.5, 0.7, 0.9\}$ and λ_2 from $\{0.1, 0.2, 0.3, 0.4, 0.5\}$. Based on the approximate number of clusters in the IDM distributions shown in Figure 4, we choose the number of interest drift levels K from $\{3, 4, 5\}$. We use an early stopping strategy with the patience set as ten times, and obtain the best-trained model based on NDGC@10 on the validation data.

5.2 Performance Comparison (RQ1 & RQ2)

The results of various methods on all datasets are reported in Table 2. Our proposed IDURL exhibits strong compatibility across the SR backbone models and significant superiority against the baselines. Moreover, we have the following observations: (1) The attention-based method SASRec consistently outperforms GRU4Rec and Caser by a large margin. It suggests the superiority of attention mechanisms on the SR problem. (2) ComiRec+ surpasses its original version, demonstrating the validity of our proposed disentanglement loss. (3) Both DuoRec and MCLRec perform better than their backbone model, *i.e.*, SASRec, which shows the effectiveness of contrastive learning (CL) techniques. DuoRec+ outperforms DuoRec, which gives credit to our stricter criterion for identifying similar users, that is, we guarantee both the same target item and interest drift degree of the positive user pairs. (4) Leveraging the category information can enhance the performance. This can be verified by comparing the results of NOVA, DIF-SR, MSSR and ASIF with that of SASRec. This phenomenon also demonstrates that category information can provide essential cues for a more comprehensive understanding of users' interests. (5) Combined with the common-used backbone model, *i.e.*, SASRec, our IDURL significantly outperforms DT4SR and STOSA, which shows the superiority of our IDURL for the interest drift issue. Moreover,

compared with other side-information integrated SR methods, our IDURL does not fuse such information into the user representations but still achieves remarkable improvements. It demonstrates the effectiveness of our manner in using multi-tag category information to learn interest drift-driven user representations.

Table 3: The results of the ablation study (Recall@10) of our IDURL with SASRec as the backbone model.

Variants	Beauty	Sports	Toys	Yelp
SASRec	0.0859	0.0494	0.0924	0.0639
w/o IDRD & IDRA	0.0863	0.0508	0.0949	0.0676
w/o IDRA	0.0882	0.0514	0.0955	0.0680
w/o IDRD	0.0906	0.0536	0.1001	0.0703
w/o IDDC	0.0883	0.0533	0.0973	0.0702
IDURL	**0.0935**	**0.0542**	**0.1012**	**0.0726**

5.3 Ablation Study of IDURL (RQ3)

We take SASRec as the backbone model of our IDURL and conduct an ablation study to analyze the contribution of the DRG, IDRD and IDRA modules. We also investigate the effect of IDDC in our prediction module by directly substituting the representation in Eq.(16) with the mean-pooling result of all drift representations.

We test the performance of our IDURL without IDDC (denoted as 'w/o IDDC'), IDURL without IDRD (denoted as 'w/o IDRD'), IDURL without IDRA (denoted as 'w/o IDRA'), IDURL without IDRD and IDRA (denoted as 'w/o IDRD & IDRA'), and IDURL without all the modules (*i.e.*, SASRec). The results are reported in Table 3.

We have the following observations: (1) **'w/o IDRD & IDRA' vs. SASRec**. The variant that retains DRG (*i.e.*, 'w/o IDRD & IDRA') beats SASRec. This means that using DRG to obtain various drift representations is beneficial for learning users' potential varying levels of interest drift. (2) **'w/o IDRA' vs. 'w/o IDRD & IDRA'**. IDURL without IDRA outperforms the variant without both IDRD and IDRA, indicating that our proposed disentanglement objective can bring more uniqueness to the generated representations, so as to optimize the potential interest drift distributions during training. (3) **'w/o IDRD' vs. 'w/o IDRD & IDRA'**. The performance gap between these two variants shows the effectiveness of constructing the contrastive views in our IDRA. And the interest drift-aware collaborative knowledge among similar users under the stricter criterion is well captured. (4) **IDURL vs. 'w/o IDDC'**. We can see that removing IDDC leads to a certain degree of performance decrease, which means that it is helpful for the item prediction module to concretize a user's potential interest drift distribution.

5.4 Hyperparameter Study of IDURL (RQ4)

We take SASRec as the backbone model to explore how different values of some key hyperparameters affect the performance.

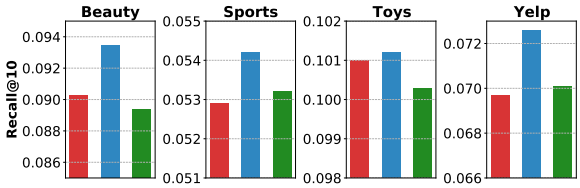
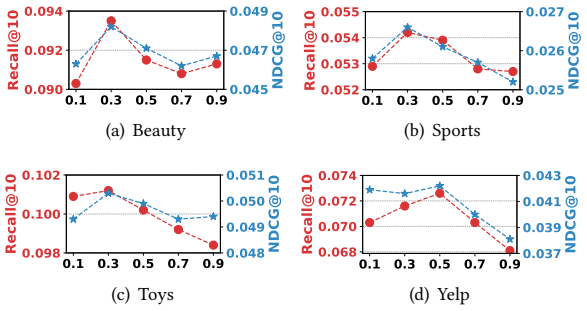


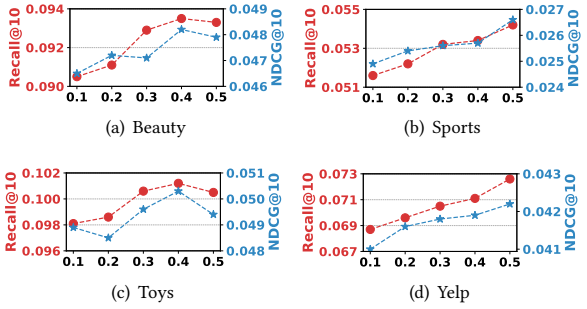
Figure 5: The performance of our IDURL with different values of K, where the values on each x-axis are 3, 4 and 5.

The number of interest drift levels K. Since the number of interest drift levels directly determines the number of generated drift representations that affects the ability to capture the interest drift, we test the performance of our IDURL with different values of $K \in \{3, 4, 5\}$ on four datasets and report the results in Figure 5. When K is equal to 4, our IDURL achieves the best performance. We can see that the performance degrades a bit when K takes a large value, which might be caused by the difficulty of disentangling too many drift representations.



(a) Beauty (b) Sports (c) Toys (d) Yelp

Figure 6: The performance of our IDURL with different values of the weight parameter λ_1. Best viewed in color.



(a) Beauty (b) Sports (c) Toys (d) Yelp

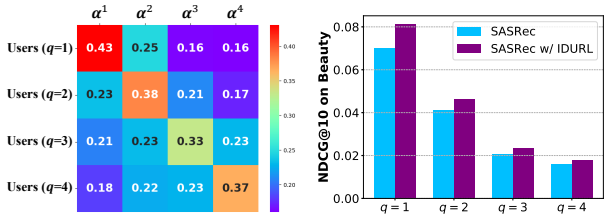
Figure 7: The performance of our IDURL with different values of the weight parameter λ_2. Best viewed in color.

Weight parameters λ_1, λ_2. We assign different values to the weight parameters λ_1 and λ_2 in Eq.(19) to study their impacts to the performance, which is shown in Figure 6 and Figure 7, respectively. On the one hand, our IDURL gets the best performance when λ_1 takes a relative small value, such as 0.3. On the other hand, a slightly larger value for λ_2 is generally more suitable for our IDURL. Moreover, we can observe that the optimal weight of $\mathcal{L}_{align}$, *i.e.*, λ_2, is often larger than λ_1. This means that the representation alignment module generally need more importance during training.

5.5 In-depth Analyses of IDURL (RQ5)

To investigate the reason why our IDURL can achieve superior performance, we select Beauty as the experimental dataset and take SASRec as the backbone model to conduct in-depth analyses on our IDURL. We find that RQ5 can be well answered by the following three merits of our IDURL.

(1) Inherent ability for concretizing users' interest drift distributions. We first set K to 4 (the optimal value from RQ4) to train our IDURL, and take the target item embedding to concretize each user's interest drift distribution via Eq.(14). Then we calculate

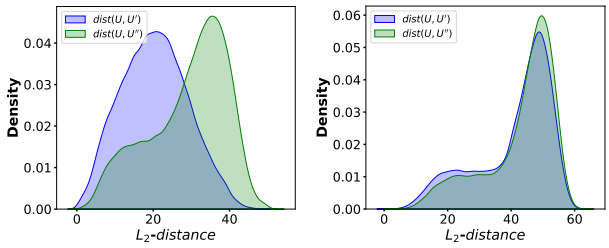


(a) (b)

Figure 8: (a) The visualization of the average distributions of interest drift levels for different user groups. (b) The performance comparison (NDCG@10 on Beauty) between SASRec and SASRec equipped with our IDURL.

each user's ground-truth interest drift degree q via IDQ module and divide all the users into 4 user groups according to their q values. Finally, we visualize the average distribution of each user group by row in Figure 8(a). We can see that the values on the diagonal are always the largest. This means that our IDURL can effectively emphasize the significance of the expected interest drift levels by assigning more prominent weights, thereby obtaining rational distributions for different user groups. It also demonstrates the effectiveness of the disentanglement objective in our IDRD.

(2) Strong performance robustness for the user groups with different interest drift degrees. Based on the results of the above grouping operation, we report the performance comparison between SASRec and SASRec equipped with our IDURL across different user groups in Figure 8(b). The improvements on every user group show that our IDURL can maintain the performance robustness for the users with different interest drift degrees.



(a) IDURL (b) IDURL w/o $\mathcal{L}_{align}$

Figure 9: The visualization of two different distributions w.r.t. the L_2-$distance$ of user representation pairs, where the representations are derived by the complete IDURL and its variant that is trained without $\mathcal{L}_{align}$.

(3) Effective representation alignment for the interest drift-aware similar users. We evaluate the alignment effect using the L_2-$distance$ of the user representation pair and visualize the effect by plotting the distance distribution via Gaussian kernel density estimation [52]. Specifically, for user u, we sample a similar user u' with the criterion of Eq.(9) and take the basic views in Eq.(10) as their respective representations, and then calculate the L_2-$distance$ of this representation pair. We repeat the above process for all the users and obtain the distance distribution w.r.t. two similar user sets, denoted as $dist(U, U')$. Then we modify the criterion for sampling similar users by requiring the same target items but ***different*** interest drift degrees. Following the same process, we obtain another distance distribution, denoted as $dist(U, U'')$. We plot the above two distributions in Figure 9(a). To better highlight the alignment effect, we also train our IDURL without $\mathcal{L}_{align}$ to obtain $dist(U, U')$ and $dist(U, U'')$, which are plotted in Figure 9(b). Compared with Figure 9(b), the distinction between the two distributions in Figure 9(a) is relatively more pronounced, and the L_2-$distance$ between users with the same interest drift degrees are generally smaller than those with different degrees (the purple distribution is much closer to the vertical axis compared with the green one in Figure 9(a)). This shows the significant alignment effect of our IDRA module. However, the two distributions in Figure 9(b) almost overlap. This verifies that when training IDURL without $\mathcal{L}_{align}$, the representation alignment among the users who share the same target items but are not interest drift-aware similar is unavoidable.

5.6 Efficiency Analyses of IDURL (RQ6)

We analyze the efficiency of our IDURL in the inference stage. Specifically, we take SASRec as the backbone model and adopt two baselines, *i.e.*, SASRec and ASIF, for the comparison of the inference time on the test data. The results are reported in Table 4. We can observe that our IDURL achieves faster inference compared with ASIF across all datasets. Meanwhile, our IDURL is slightly slower than SASRec, which is expected, because it takes SASRec as the backbone model and needs other computations about drift representations. The results demonstrate that our IDURL remains competitive in terms of inference speed compared with the baselines.

Table 4: The results of inference times of two baselines and our IDURL on the test data of four datasets.

Datasets	Beauty	Sports	Toys	Yelp
ASIF	2.091s	2.602s	1.699s	2.854s
IDURL	1.459s	1.790s	1.064s	1.834s
SASRec	1.283s	1.398s	0.929s	1.427s

6 Conclusions and future work

In this paper, we highlight that most SR methods, which adopt the common paradigm of user representation learning, often encounter two challenges arising from users' interest drift. To enhance these SR methods and address the challenges, we propose a novel framework named IDURL. In IDURL, we propose an IDQ module to quantitatively measure the interest drift. Moreover, we design a DRG module to model users' various levels of interest drift, and an IDRD module that optimizes the potential distributions of the interest drift under the guidance of IDQ. Furthermore, we introduce an IDRA module to capture the interest drift-aware collaborative knowledge among users. Finally, we obtain the users' overall interest representations and calculate their preference scores, based on the concretized interest drift distributions. Extensive experiments on four public datasets show the effectiveness of our IDURL.

In the future, we are interested in leveraging the rich knowledge of large language models to investigate the interest drift issue.

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